

13-01-2026

Lab - 8

Step 1: Import Libraries

This step imports all necessary libraries for data processing, visualization, and machine learning.

```
In [ ]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

Step 2: Load the Dataset

Load Given dataset - heart.csv

```
In [ ]: df = pd.read_csv('heart.csv')
```

Step 3: Data Overview

In this step, we examine the dataset structure, summary statistics, and check for missing values.

```
In [ ]: df.info()
```

```
In [ ]: df.describe()
```

Step 4: Univariate Analysis

Here we visualize the distribution of each feature using histograms.

```
In [ ]: df.hist(figsize=(12,12))  
plt.show()
```

Step 5: Bivariate Analysis

This step involves exploring the correlations between features using a heatmap.

```
In [ ]: %matplotlib inline
```

```
In [ ]: sns.heatmap(df.corr(),annot=True)
```

Step 6: Outlier Detection

We visualize potential outliers using boxplots.

```
In [ ]: df.boxplot()
```

Step 7: Split Data into Training and Testing Sets

The dataset is split into training and testing sets for model evaluation.

```
In [ ]: x = df.drop('target',axis=1)
```

```
In [ ]: y = df['target']
```

```
In [ ]: x.head()
```

```
In [ ]: y.head()
```

```
In [ ]: from sklearn.model_selection import train_test_split
```

```
In [ ]: x_train, x_test, y_train, y_test = train_test_split(x,y,test_size=0.3,random_state=
```

Step 8: Train Decision Tree

We train a Decision Tree Classifier on the training data. You have to also check for KNeighborsClassifier and GaussianNB

```
In [ ]: from sklearn.tree import DecisionTreeClassifier
```

```
In [ ]: model = DecisionTreeClassifier()

In [ ]: model.fit(x_train,y_train)

In [ ]: from sklearn.metrics import accuracy_score

In [ ]: y_pred = model.predict(x_test)

In [ ]: model.score(x_train,y_train)

In [ ]: model.score(x_test,y_test)

In [ ]: accuracy_score(y_test,y_pred)
```

Cross Validation

```
In [ ]: from sklearn.model_selection import KFold

In [ ]: k = KFold(n_splits=10)

In [ ]: x

In [ ]: for train_index, test_index in k.split(x):
    x_train, x_test = x.iloc[train_index], x.iloc[test_index]
    y_train, y_test = y.iloc[train_index], y.iloc[test_index]
    model_1 = DecisionTreeClassifier()
    model_1.fit(x_train, y_train)
    print(model_1.score(x_test, y_test))

In [ ]: from sklearn.model_selection import cross_val_score

In [ ]: cross_val_score(model,x,y,cv=10)

In [ ]: import numpy as np

In [ ]: x1 = np.arange(1,11).reshape(-1,1)

In [ ]: y1 = np.array([0,1,0,1,0,1,0,1,0,1])

In [ ]: for i,j in k.split(x1):
    print("<----->")
    print(x1[i])
    print(x1[j])
    x_train,x_test = x1[i],x1[j]
    y_train,y_test = y1[i],y1[j]
```